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End Semester Examination December 2024

Name of the Course: B.Tech (CSE)

Semester: III

Name of the Paper: Fundamental of AI and ML

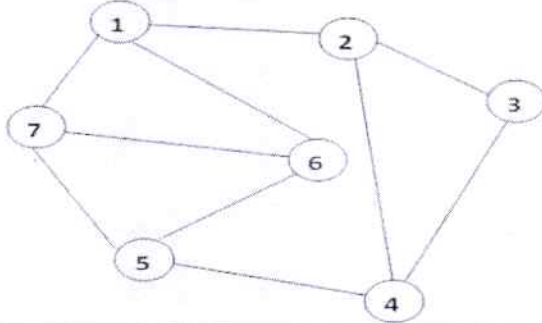
Paper Code: TCS 364

Time: 3 Hours

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20 marks)	CO1 CO2																								
(a)	What is the goal of AI? Explain advantages and disadvantages of AI.																									
(b)	What is searching? Explain different types of searching algorithm used in AI.																									
(c)	Compare Uninformed search method and Informed search method. Find the path to reach the goal node (4) from the star node (1) by applying DFS. 																									
Q2	(20 marks)	CO2 CO3																								
(a)	(i) Show that: $(p \Rightarrow (q \Rightarrow r)) \Rightarrow ((p \Rightarrow q) \Rightarrow (p \Rightarrow r))$ is a tautology without using truth tables. (ii) Show that $[(p \Rightarrow q) \wedge (q \Rightarrow r)] \Rightarrow (p \Rightarrow r)$																									
(b)	Consider the following training set having three different attributes namely Brightness and Saturation: <table border="1" data-bbox="596 1536 1145 1798"> <thead> <tr> <th>Brightness</th><th>Saturation</th><th>Class</th></tr> </thead> <tbody> <tr><td>40</td><td>20</td><td>Red</td></tr> <tr><td>50</td><td>50</td><td>Blue</td></tr> <tr><td>60</td><td>90</td><td>Blue</td></tr> <tr><td>10</td><td>25</td><td>Red</td></tr> <tr><td>70</td><td>70</td><td>Blue</td></tr> <tr><td>60</td><td>10</td><td>Red</td></tr> <tr><td>25</td><td>80</td><td>Blue</td></tr> </tbody> </table> Using the K-nearest algorithm (KNN) algorithm with K=3, Classify the following points (i) (20,35) (ii) (15,30)	Brightness	Saturation	Class	40	20	Red	50	50	Blue	60	90	Blue	10	25	Red	70	70	Blue	60	10	Red	25	80	Blue	
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(c)	Differentiate between Markov decision processes and Bayesian networks. Explain with suitable examples.																									

Q3	(20 marks)			CO3, CO4																									
(a)	Differentiate between Supervised, Unsupervised and Reinforcement machine learning?																												
(b)	Use the k-means algorithm and Euclidean distance to cluster the following example into 2 clusters: <table><tr><th>Height</th><th>Weight</th></tr><tr><td>185</td><td>72</td></tr><tr><td>170</td><td>56</td></tr><tr><td>168</td><td>60</td></tr><tr><td>179</td><td>68</td></tr><tr><td>182</td><td>72</td></tr><tr><td>188</td><td>77</td></tr><tr><td>180</td><td>71</td></tr><tr><td>180</td><td>70</td></tr><tr><td>183</td><td>84</td></tr><tr><td>180</td><td>88</td></tr><tr><td>180</td><td>67</td></tr><tr><td>177</td><td>76</td></tr></table>				Height	Weight	185	72	170	56	168	60	179	68	182	72	188	77	180	71	180	70	183	84	180	88	180	67	177
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177	76																												
(c)	What are the different methods for measuring classifier performance? Based on the table below, Calculate Classification Accuracy, Precision, Recall, F1-score, and Specificity. <table><tr><td>n = 100</td><td>Actual: No</td><td>Actual: Yes</td><td></td></tr><tr><td>Predicted: No</td><td>TN: 65</td><td>FP: 3</td><td>68</td></tr><tr><td>Predicted: Yes</td><td>FN: 8</td><td>TP: 24</td><td>32</td></tr><tr><td></td><td>73</td><td>27</td><td></td></tr></table>			n = 100	Actual: No	Actual: Yes		Predicted: No	TN: 65	FP: 3	68	Predicted: Yes	FN: 8	TP: 24	32		73	27											
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	73	27																											
Q4	(20 marks)			CO4, CO5																									
(a)	The value of independent variable X and dependent value of Y are given below: <table><tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Y</td><td>3</td><td>4</td><td>6</td><td>5</td><td>7</td></tr></table> Find the mean square regression and minimum cost function of the line $Y=wX+b$. Estimate the value of Y when X is 7.				X	1	2	3	4	5	Y	3	4	6	5	7													
X	1	2	3		4	5																							
Y	3	4	6	5	7																								
(b)	Difference between Data Pre-processing and Data Cleaning. What are the steps involved in preprocessing data? Why is preprocessing important in data?																												
(c)	What is a decision tree? Discuss the use of decision tree for classification purpose with an example.																												
Q5	(20 marks)			CO5, CO6																									
(a)	Describe regression and classification with an example. Explain how to implement these techniques using the sklearn library.																												
(b)	Describe bagging and boosting ensemble learning with example. What is Random Forest? Random forest is an example of which ensemble learning.																												
(c)	Explain the following: 1. Mean and Mode 2. Gradient Descent 3. Logistic regression cost function																												